

Can Native MXFP4 Replace BF16 in FPGA Gated DeltaNet Decode?

Cost and Long-Horizon Recurrent-State Drift

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Abstract—This paper asks whether native Open Compute Project (OCP) MXFP4 arithmetic can replace bfloat16 (BF16)-style arithmetic in a persistent-state Gated DeltaNet recurrence-core kernel on an AMD Alveo U55C field-programmable gate array (FPGA). Persistent state and the five-phase token schedule are inherited from prior work and held fixed. We implement E2M1 products in integer lookup-table (LUT) logic and E8M0-aware accumulation, then separate floating quantize/dequantize (Q/DQ) diagnostics from the native encoded path. At token 8,192, BF16 gives 0.999955 output cosine and 0.008988 recurrent-state relative L_2 error against single-precision floating point (FP32). MXFP4 Q/DQ gives 0.729275 and 1.103841, whereas the native encoded path gives 0.429834 and 2.580730. The encoded HLS kernel uses 1.636 times the LUTs and 1.437 times the maximum estimated STEP cycles of the fixed-geometry BF16 kernel. A recurrence-aware residual/write-log candidate passes 3 test seed blocks crossed with two initial-state conditions and two 8,192-token development conditions, but uses 5.124 times the BF16 LUTs and misses 3 explicit $\text{II}=1$ constraints. Thus the present native MXFP4 implementations do not support replacing BF16. The corrected candidate’s out-of-context post-route image fits using 208,523 CLB LUTs and 624 of 960 URAMs, but has -1.540 ns worst negative slack at 250 MHz; the first passing point in the tested fixed-route sweep is 180.18 MHz. The long-horizon quality result is layer-level and synthetic. Four short model-derived Qwen layer-12 recurrence traces preserve the same arithmetic ordering but contain only 12–18 valid tokens each. Power is a vectorless estimate rather than board energy.

Index Terms—Gated DeltaNet, MXFP4, recurrent state, FPGA, numerical drift, Vitis HLS

I. INTRODUCTION

Gated DeltaNet (GDN) replaces a growing attention history with a fixed-size recurrent matrix that is decayed and updated once per token [1]. This makes the numerical state persistent: an arithmetic error written at one token can affect every later token. Lowering state precision therefore changes more than storage cost. It changes the dynamical system implemented by the decoder.

Gupta et al. map GDN decode to a field-programmable gate array (FPGA) using persistent on-chip state and a five-phase BF16 pipeline [2]. We use that organization as the baseline dataflow. Neither persistent state nor the five-phase schedule is a contribution of this work. We study only the arithmetic replacement inside that recurrence-core boundary.

The Open Compute Project (OCP) MXFP4 format stores 32 E2M1 elements with one E8M0 scale [3]. Its 4.25 logical bits per value are attractive for a recurrent state that is read and rewritten every token. The shared exponent also creates

two risks. Values below the block range disappear during alignment, and a scale change requantizes the complete block. Those errors feed back through the next state update.

We evaluate the replacement as a controlled experiment. The recurrence, logical K-by-V state, layer dimensions, U55C target, 4.0 ns constraint, $P_K = 16$, $P_V = 8$, and block size 32 are fixed. The study contributes:

- 1) an encoded-integer MXFP4 recurrence oracle and native high-level synthesis (HLS) arithmetic with explicit E2M1 decoding, E8M0 alignment, round-to-nearest-even (RNE), Q1.15 controls, and event counters;
- 2) long synthetic traces at 64, 256, 1,024, 4,096, 8,192 tokens for FP32, BF16, MXFP4 floating Q/DQ, native encoded MXFP4, flat INT4, and a floating-Q/DQ MXFP8-state fallback, plus four short model-derived recurrence traces and 6 scale refresh policies;
- 3) a fixed-geometry BF16, encoded-MXFP4, and native-MXFP8 HLS comparison and a separately classified recurrence-aware correction, including out-of-context post-route timing, utilization, design-rule, and vectorless-power evidence for the correction and MXFP8 baseline; and
- 4) an evidence-backed negative answer for the current implementations, including the numerical failure of native encoded MXFP4 and the hardware cost of the correction that meets the frozen long-horizon synthetic gate.

The evaluation covers one recurrence core, synthetic recurrent traces, four short model-derived layer-12 recurrence traces, matched HLS estimates, and out-of-context implementation attempts. It does not measure Qwen3-Next perplexity, downstream accuracy, or board energy. Within that boundary, can native MXFP4 replace BF16-style arithmetic while reducing FPGA cost and limiting empirical recurrent-state drift over long synthetic decode traces?

II. BACKGROUND AND SCOPE

A. Microscaling Representation

An E2M1 element contains 1 sign bit, 2 exponent bits, and 1 mantissa bit. Its positive finite magnitude codebook is $\{0, 0.5, 1, 1.5, 2, 3, 4, 6\}$. Both signed-zero codes decode to zero, and the encoder emits positive zero. An E8M0 byte encodes the shared power-of-two scale 2^{e-127} for codes 0 through 254. Code 255 is invalid; an all-zero block uses scale code 127. NaN and infinity inputs are rejected. These rules

follow the OCP representation [3]; scale selection and RNE at each state-write boundary are implementation choices.

B. Recurrence-Core Boundary

The head counts and K-by-V dimensions follow the pinned official Qwen3-Next-80B-A3B-Instruct configuration and Transformers implementation [4], [5]. For value head h , query/key head $p(h) = \lfloor h/2 \rfloor$ serves the corresponding pair of value heads. With logical state $S_h \in \mathbb{R}^{K \times V}$ and $K = 128$, $V = 128$, the token recurrence is

$$S_h^d = \alpha_h S_h, \quad (1)$$

$$u_h = \beta_h \left(v_h - k_{p(h)}^\top S_h^d \right), \quad (2)$$

$$S_h' = S_h^d + k_{p(h)} u_h^\top, \quad (3)$$

$$o_h = q_{p(h)}^\top S_h'. \quad (4)$$

The query is normalized and scaled by $1/\sqrt{128}$ before entering the kernel; the key is normalized. Gated RMSNorm, the z branch, projections, depthwise convolution, and output projection are outside the measured boundary. We therefore call the implementation a GDN recurrence-core kernel, not a complete GDN layer. Holding the unchanged convolution, projections, and output normalization outside the boundary isolates arithmetic-induced drift and cost in the persistent recurrence rather than mixing them with unmodified layer components.

The operation order is decay, prediction from the decayed state, delta formation, rank-one write, and output from the updated state. The software oracle was checked against pinned Transformers and FLA recurrent, chunk, and cache paths before quantized experiments were admitted [5], [6].

C. Prior Work and Component Ownership

DeltaNet establishes the rank-one erase/write recurrence and parallel training formulations on which GDN builds [7]; the gated recurrence itself is inherited from Yang et al. [1]. The persistent-state FPGA organization, paired-head handling, and five phases are inherited from Gupta et al. [2]. MXFP4-B32 is standardized [3]. QLLama maps microscaling quantization to FPGA inference, while Jack of All Scales studies MXFP dot-product mappings and digital signal-processing (DSP) block support on Agilex-5 [8], [9]. MXFormer implements block-exponent-aligned MXFP4 compute in charge-trap memory, MicroMix mixes MXFP4/6/8 channels, and MXAttention and an MXFP post-training-quantization benchmark identify scaling, clipping, and underflow as material quality factors [10], [11], [12], [13]. These are precedents for alignment, mixed precision, and scale-policy studies. Zhang et al. quantize the chunk-wise GDN matrix-inversion path rather than token-serial recurrent state [14]. KIVI retains recent full-precision entries beside a quantized cache, and SSDi8 uses a persistent INT8 state-space path with error correction [15], [16]. These works establish prior art for the component ideas. Our scope is the matched arithmetic-replacement experiment and its verification, not the invention of those constituents. Table I makes the resulting ownership boundaries explicit.

TABLE I
COMPONENT OWNERSHIP RELATIVE TO THE CORRECTED IMPLEMENTATION.

Component	Classification	Basis
Persistent on-chip state	Inherited	Gupta et al. [2]
Five-phase token schedule	Inherited	Gupta et al. [2]
Qwen recurrence and core boundary	Corrected	Pinned software parity
OCP MXFP4-B32 state	Modified	Standard format in a recurrent path
E2M1/E8M0 FPGA arithmetic	Modified	OCP format plus QLLama and Jack of All Scales [3], [8], [9]
Encoded arithmetic oracle	Newly proposed	Artifact-specific independent integer implementation
E2M1/E2M0 residual plus fixed write log	Modified	Related residual-buffer methods exist [15], [16]; no first-use claim
Long-horizon drift harness	Newly proposed	Same recurrence and synthetic input prefix

Note: Newly proposed identifies ownership within this artifact, not a claim of literature priority.

III. NUMERICAL METHOD

A. Uniform MXFP4 Path

The matched BF16 comparator rounds state and operands to BF16 at their declared boundaries and performs reductions in FP32. The MXFP8-state fallback changes only the recurrent state to eight-bit E4M3 elements under E8M0 B32 scales while retaining MXFP4 token operands; flat INT4 uses signed symmetric four-bit quantization with a tensor-wide scale.

We report a floating Q/DQ MXFP4 diagnostic and a distinct native encoded path. The Q/DQ diagnostic quantizes at named boundaries but performs the intervening recurrence in floating point. It does not represent the HLS accumulator or its event behavior. The encoded path below is the software counterpart of the native HLS arithmetic.

The uniform path stores E2M1 elements in B32 blocks with an E8M0 scale per block. For nonzero block maximum m and largest finite E2M1 magnitude $x_{\max}$, the selected scale power is $s = \lceil \log_2(m/x_{\max}) \rceil$; its biased E8M0 code is clamped to the valid range. Elements are divided by 2^s and rounded to E2M1 with RNE. Zero has a canonical encoding. If E2M1 magnitude code r denotes $r/2$, a product has integer magnitude $r_a r_b$ and power $s_a + s_b - 2$. One flat 128-term reduction aligns every product directly to the greatest nonzero product exponent (power), applies RNE to the full exponent-difference shift, and accumulates in ascending key-index order with signed 32-bit saturation after every addition. A nonzero term rounded to zero is counted as an alignment underflow. This uniform baseline has no guard-bit offset; the five guard bits described below belong only to the corrected candidate.

Each synthetic STEP record supplies normalized q and k , value v , decay α , delta gate β , and the selected initial state S . The MXFP4 paths encode q , k , v , and every state write; α and β use unsigned Q1.15 fixed point. There are no projection weights in a test vector because all projections are outside the recurrence-core boundary. The MXFP8 fallback changes only state element codes to E4M3 under B32 E8M0 scales;

flat INT4 changes the same token operands and state under its declared tensor-wide scale. Requantization, element and accumulator saturation, scale clamps, alignment underflows, and scale changes are explicit state-write events. The complete contract is versioned in the artifact rather than inferred from C++ types.

The element product uses sign exclusive-or (XOR), table-based E2M1 magnitude multiplication, and exponent addition. No floating-point type appears in the active native arithmetic sources. The resulting E2M1 element-product logic is integer based; wider coefficient operations may still map to DSPs and are reported as such.

B. Recurrence-Aware Correction

Native encoded MXFP4 fails the frozen finite-trace thresholds, so we also evaluate one locally frozen correction. The base state has an E2M1 primary term and a dense signed 3-bit E2M0-style residual, each with B32 scales. 2-term residual-stacked E2M1 vectors represent token inputs and live rank-one writes. Up to 7 writes remain in a log with Q1.15 decay coefficients. At capacity, all 7 entries are folded tilewise into the base and the log is cleared. This candidate aligns shared exponents with 5 guard bits before signed 32-bit accumulation.

This representation is a mitigation, not the controlled uniform replacement. It changes the physical state representation and adds a periodic fold. Its logical payload is 537,744 bytes per layer, only 2,928 bytes below a uniform MXFP8-E4M3-B32 state. Exact arithmetic equivalence was tested separately before quantized evaluation.

IV. FPGA IMPLEMENTATION

Both controlled baselines use equivalent persistent-state command semantics and the same five-phase token organization. Commands select one of 36 runtime layer slots and support RESET, LOAD, STEP, and READBACK. Their data-plane encodings are not identical: BF16 transports BF16 tensors, whereas MXFP4 transports encoded elements and scales and returns expanded integer output fields. Packing and token quantization are outside both HLS kernels. Before interface packing, a STEP consumes 16,512 logical input bytes in the BF16 kernel and 4,480 in the uniform MXFP4 kernel; the corrected residual-stack candidate consumes 8,832. BF16 returns 8,192 logical output bytes, while the native kernels' expanded INT32-mantissa/INT16-exponent fields total 24,576. These are type-layout counts, not measured AXI traffic; RESET has no tensor payload, and resident-state LOAD and READBACK use their separately reported logical state payloads. The HLS target is xcu55c-fsvh2892-2L-e under Vitis HLS 2025.2 at 4.0 ns with $P_K = 16$, $P_V = 8$, and B32. We retain the U55C target used by the inherited BF16 dataflow so the arithmetic comparison does not also change devices. We fix this target and parallelism point for the replacement rather than tune each format independently.

The corrected candidate adds packed primary and residual banks, 7 packed key/update log entries, per-head gamma and lambda coefficients, and event counters. A non-mutating

snapshot exposes all persistent fields for exact verification. Folding is atomic at the command boundary.

Figure 1 maps the correction onto the inherited five-phase schedule. The base state remains resident while each STEP decays its coefficient, evaluates the base and live writes, appends the new rank-one update, and reads the output from the updated logical state. When the 7-entry log fills, a tilewise fold runs after the output has observed the newest write.

Logically, B32 blocks run along the V-column axis within each K row. The code partitions the K and V access lanes by P_K and P_V and targets an initiation interval (II) of 1 on inner loops. Source directives request block RAM and UltraRAM bindings, but the inferred memory counts contradict bit-capacity lower bounds for the declared deep arrays and are excluded from paper conclusions. After removing invalid pipeline directives from long validation and requantization loops, HLS estimates a 3.108 ns path. The 4.0 ns target and tool-reported uncertainty imply a 2.920 ns effective budget, so the estimate misses the configured margin by 0.188 ns. 3 remaining targeted loops achieve II=2 rather than II=1. HLS reports 78 DSPs for coefficient and wider arithmetic; the E2M1 element multiplier itself remains integer logic. These source bindings and estimates do not establish a physically allocated 36-layer bank by themselves; Section VI-D separately reports the corrected candidate's out-of-context implementation.

V. EVALUATION

A. Questions and Controls

The experiments answer three questions:

- Does native encoded MXFP4 limit empirical output and recurrent-state drift over 8,192 synthetic tokens relative to FP32 and BF16?
- Do floating-Q/DQ scale refresh policies or an MXFP8-E4M3 state fallback meet the same finite-trace criteria?
- If a recurrence-aware correction meets the finite-trace thresholds, does its HLS estimate improve on the fixed-geometry BF16 kernel?
- Does that corrected candidate physically fit and close the declared clock target after out-of-context routing?

The controlled nominal baseline variants use 16 query/key heads, 32 value heads, $K = 128$, $V = 128$, the same K-by-V state, and the same generated input prefix. The development seed is 0xFB72. Required checkpoints are 64, 256, 1,024, 4,096, 8,192. The nominal trace draws query and key vectors from independent standard normal distributions before normalization. Values and random initial states are zero-mean normal with standard deviations 0.25 and 0.05, respectively. Nominal decay and delta gates follow $\mathcal{U}[0.950, 1.000]$ and $\mathcal{U}[0.000, 1.000]$. The high-retention correction traces keep the vector distributions but use $\alpha \sim \mathcal{U}[0.995, 1.000]$ and $\beta \sim \mathcal{U}[0.850, 1.000]$, with both random and zero initial states. Changing the trace-family tag changes the deterministic RNG stream, even at the same seed. The correction results are therefore not paired numerical ablations of the nominal baseline trace. The frozen engineering gate requires output

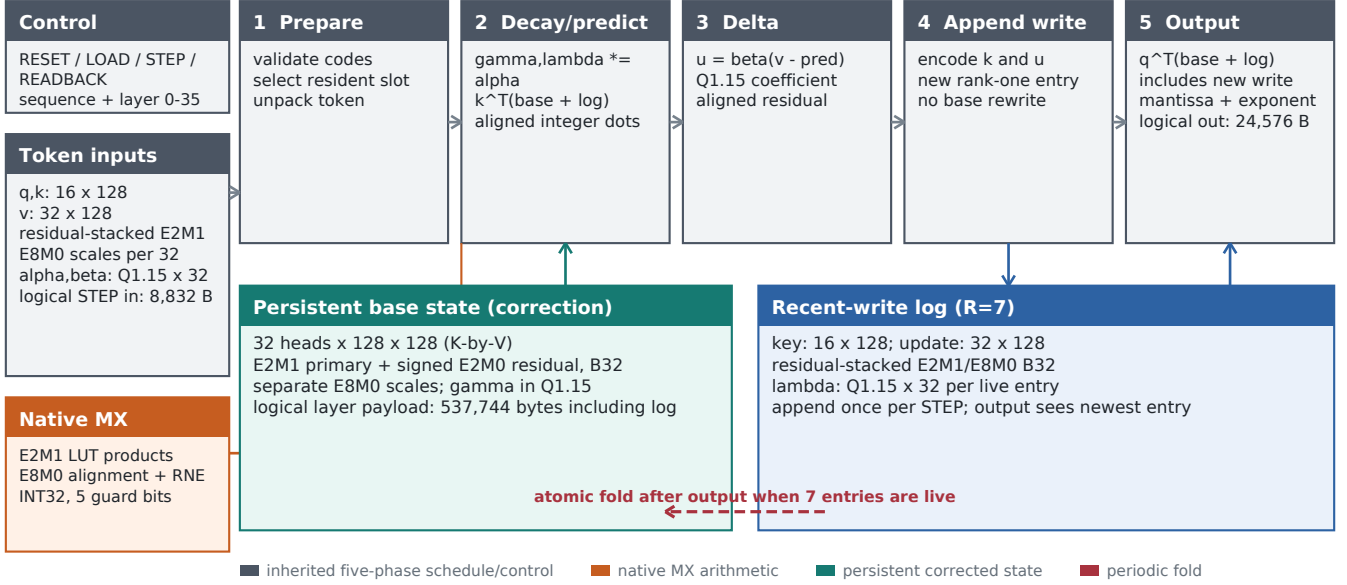


Fig. 1. Corrected-candidate datapath on the inherited five-phase token schedule. Gray denotes inherited control and scheduling, orange denotes native MX arithmetic, teal and blue denote the corrected persistent representation, and the dashed path is the periodic atomic fold. Dimensions and rates are generated from the numerical contract; the drawing is logical architecture, not physical-placement evidence.

cosine at least 0.99 at every checkpoint, final state relative L_2 at most 0.10, and zero element saturation, accumulator saturation, and scale-clamp events. These are the three hard-event counters; alignment underflows and deliberate E2M0 residual clips are reported but excluded from the zero-event condition. We fixed these complementary diagnostics before test-set access: cosine constrains observable output direction, while state relative L_2 constrains hidden-state magnitude. They were not calibrated to downstream model quality.

The correction uses 3 test seed blocks crossed with random and zero initial states, producing 6 paired 1,024-token conditions. Every artifact was deterministically recomputed by the same implementation and compared field by field with its recorded comma-separated values (CSV) artifact. This checks integrity and determinism, not implementation independence. The 2 8,192-token development runs use the same recorded seed with random and zero initial states. Each longer run was fully recomputed from source and compared field by field with its recorded CSV artifact. The encoded HLS candidate was also checked for 267 frozen arithmetic cases and for 64 stateful C-simulation tokens through repeated folds and final snapshot readback.

The corrected candidate is synthesized and routed out of context for the declared U55C part. A source-level clock-buffer insertion is hash-checked against the HLS-generated top before implementation. Timing is re-reported on the same routed checkpoint at several clock periods; the first passing tested point is not interpreted as a binary-searched maximum frequency. Post-route power uses Vivado vectorless activity propagation at a timing-closed 5.6 ns period. It is a tool

estimate, not board telemetry. Vitis HLS/XSIM also runs 2 exact early-return commands through the generated RTL. Those commands exercise control/status behavior but no recurrent state transition. A direct AXI harness over the same HLS-generated Verilog additionally completes an exact candidate LOAD in 4,797,322 cycles. Two isolated official 64-token XSIM attempts exhaust host memory before transaction one, so neither result is reported as candidate recurrent RTL parity.

Separately, a pinned real Qwen3-Next layer-input capture at layer 12 is combined with the matching checkpoint projection weights to reconstruct query, key, value, decay, and delta-gate inputs at the post-convolution, pre-query/key-normalization recurrence boundary. The 4 prompt traces contain 60 valid tokens in total, with 12–18 tokens per prompt. They are evaluated as floating Q/DQ diagnostics and are not mixed into the synthetic long-horizon curves. This is a short real-input recurrence check, not closed-loop quantized model inference.

B. Metrics and Evidence Levels

Output cosine similarity flattens all value heads and V coordinates at each token. State relative L_2 flattens all heads, K rows, and V columns and is $\|S_q - S_{32}\|_F / \max(\|S_{32}\|_F, 1 \times 10^{-12})$; maximum absolute state error and arithmetic events are also recorded. Exact equality returns cosine one and zero error; the artifact specifies the near-zero-norm cosine convention. No headwise minimum, percentile, or acceptance gate is computed. Software traces, HLS C simulation, HLS synthesis, register-transfer-level (RTL) simulation, post-route implementation, and board measurement are separate evidence levels. A PASS at one level does not fill a missing later level.

VI. RESULTS

A. Uniform Arithmetic Replacement

Figure 2 and Table III show the controlled synthetic trace. BF16 remains close to FP32 through the final checkpoint. Both MXFP4 Q/DQ and native encoded MXFP4 fail the frozen criteria, and the encoded path drifts more than its floating diagnostic. Flat INT4 also fails. The floating-Q/DQ MXFP8-E4M3 state fallback improves on four-bit state but still misses both final criteria. Thus the Q/DQ curve cannot be paired with the encoded HLS cost as if they were one implementation.

Two supplemental full-geometry, 8,192-token Q/DQ traces test deterministic exponent cycling and alternating-sign cancellation. Each pair below reports (output cosine, state relative L_2) at token 8192. Dynamic range gives BF16 (0.999875, 0.016476), MXFP4 (0.410379, 26.772181), and MXFP8 state (0.698070, 2.314652). Cancellation gives BF16 (0.999997, 0.008844), MXFP4 (0.981307, 1.667123), and MXFP8 state (0.997011, 0.255859). BF16 stays within both full-trace thresholds; every low-precision variant fails at least one, although MXFP8 state is the strongest low-precision comparator. These traces are synthetic range/cancellation probes, not model-derived outlier evidence.

a) *Real-capture checks.*: The pinned Qwen layer-input capture contains 147,456 values. Its absolute-value 99th and 99.9th percentiles are 0.6836 and 2.5781, while the maximum is 51.25, or 75.0 times the 99th percentile. This confirms that a real layer boundary can contain a much wider tail than the nominal Gaussian trace. The reconstructed recurrence traces in Table II give BF16 mean final output cosine/state relative L_2 of 0.999991/ 0.005894, MXFP4 gives 0.982048/ 0.324788, MXFP8 state gives 0.989173/ 0.207192, and INT4 gives 0.866071/ 0.774523. The ordering agrees with the synthetic study, but these short traces cannot establish long-horizon stability, perplexity, or downstream accuracy.

No tested floating-Q/DQ scale policy rescues uniform MXFP4. Fixed scales retain more output direction but produce extensive state-element clipping; frequent refresh reduces clipping without satisfying the drift gate. Periodic and threshold-triggered refresh also fail. Table IV reports the complete token-8,192 ablation rather than selecting one favorable metric.

B. Fixed-Geometry HLS Cost

Table V holds the recurrence, dataflow, target, parallelism, and block size fixed. The native encoded MXFP4 point is larger and has a longer estimated STEP schedule than BF16, so the HLS cost hypothesis fails. Native MXFP8 is also synthesized at the same boundary as a wider low-precision baseline. The different data-plane encodings and excluded packing mean this is not an end-to-end interface or energy ratio. All inferred memory counts are omitted because they conflict with bit-capacity lower bounds for the declared arrays.

Figure 3 places the evidence levels side by side without filling missing measurements. The declared 36-core BF16 state is 36.00 MiB and requires an ideal minimum of 1,024 URAM288 primitives, already above the U55C total of 960.

Uniform MXFP4 is 9.56 MiB and has an ideal lower bound of 256 URAM288 plus 256 BRAM18K, so it passes only the raw-bit necessary condition. These totals cover one sequence of recurrence-core state; they exclude weights, convolution state, full-attention key/value caches, activations, ports, banking, replication, scratch space, placement, and routing. Consequently, they do not establish which complete model fits on the card. The matched all-layer BF16 synthesis requires 9,216 RAMB36/FIFO primitives versus 2,016 available, so physical fit is FAIL and no BF16 routed timing or power result exists. A reduced-layer design is not substituted because it would change the controlled state layout. The HLS panel contains matched BF16, native encoded MXFP4, and native MXFP8 kernels; matched INT4 remains unrun. No energy axis is shown because energy was not measured.

C. Recurrence-Aware Correction

The correction passes all 3 test seed blocks under both initial-state conditions. Table VI reports the worst of the 6 paired test conditions beside the two fully recomputed 8,192-token development conditions. Passing the three-hard-counter gate does not imply event-free arithmetic: the longer runs contain large alignment-underflow and deliberate residual-clip counts. The seed block, not each initial-state condition, is the stochastic unit. These are finite-trace engineering checks, not a lossless-arithmetic or robustness claim.

The numerical recovery does not produce an FPGA Pareto improvement. Table VII compares logical storage and HLS estimates with both baselines. The correction is larger and slower even before physical implementation; its periodic fold also creates a large command burst. Three explicit $\Pi=1$ constraints fail.

D. Post-Route and Generated-RTL Evidence

The corrected candidate physically fits the U55C in the out-of-context flow. It uses 208,523 CLB LUTs, 111,027 flip-flops, 353 BRAM tiles, 624 URAMs, and 44 DSPs. URAM is the limiting reported resource at 65.0 percent. The implementation has 38 design-rule warnings but no critical warnings or errors.

Physical fit does not imply target timing. At 250 MHz the design has -1.540 ns WNS, and the 200-MHz check also fails. The first passing point in the tested fixed-route sweep is 180.18 MHz. The 4.971 ns target-clock critical path is 92.5 percent routing with 5 logic levels and terminates at a resident-state URAM reset/write-enable pin. It is not an E2M1 multiplier path.

Official Vitis HLS/XSIM co-simulation passes 2 of 2 generated-RTL control commands at 19,971 cycles per command. Both commands return before a recurrent update, so the candidate's 64-token exact recurrent result remains C simulation, not RTL parity. The direct Verilator snapshot LOAD is PASS, but the harness completes 0 recurrent STEPs; the required 64-token candidate result is therefore NOT_ESTABLISHED. Table VIII preserves this boundary.

At a timing-closed 5.6 ns period, Vivado reports 6.991 W total on-chip power: 3.582 W dynamic and 3.410 W static,

TABLE II

FLOATING-Q/DQ RECURRENCE DIAGNOSTICS ON FOUR MODEL-DERIVED QWEN LAYER-12 PROMPT TRACES. FINAL VALUES ARE AVERAGED PER PROMPT; “WORST” AND MAXIMUM ABSOLUTE ERROR RANGE OVER ALL PROMPTS AND VALID TOKEN POSITIONS.

Arithmetic	Mean final cos.	Worst cos.	Mean state L_2	Worst state L_2	Max abs.	Nonfinite
BF16 state/operands	0.99999	0.99998	0.00589	0.00657	0.01917	0
MXFP4 state/operands	0.98205	0.96768	0.32479	0.33888	0.61595	0
MXFP4 operands + MXFP8 state	0.98917	0.96967	0.20719	0.21994	0.46683	0
Flat INT4	0.86607	0.75192	0.77452	0.82253	0.40506	0

Long-horizon synthetic recurrence drift

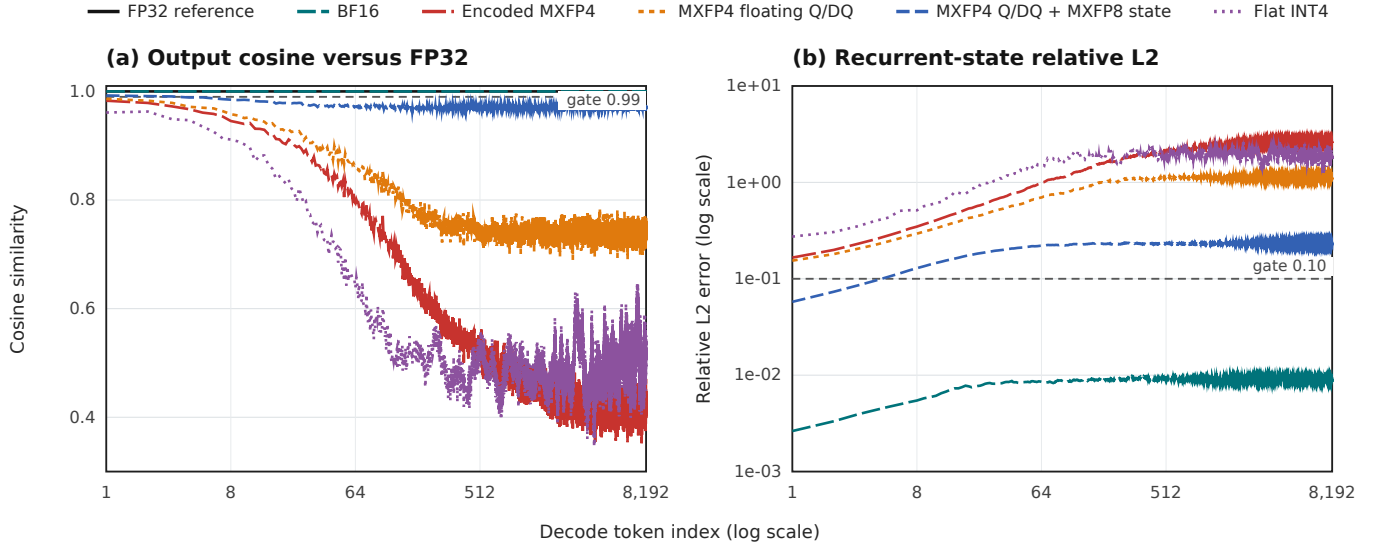


Fig. 2. Layer-level synthetic output cosine (left) and recurrent-state relative L_2 error (right) versus token index. The horizontal threshold guides are engineering criteria, not model-quality guarantees. Each line is one deterministic nominal development trace. Curves distinguish floating Q/DQ from native encoded MXFP4; one seed cannot estimate trace-population uncertainty.

TABLE III

CONTROLLED SYNTHETIC DRIFT AT EVERY REQUIRED CHECKPOINT. EACH CELL GIVES OUTPUT COSINE AND STATE RELATIVE L_2 AGAINST FP32.

Variant	64		256		1,024		4,096		8,192	
	Cos.	Rel. L_2	Cos.	Rel. L_2	Cos.	Rel. L_2	Cos.	Rel. L_2	Cos.	Rel. L_2
BF16 state/operands	0.99996	0.00853	0.99996	0.00905	0.99996	0.00873	0.99996	0.00898	0.99996	0.00899
MXFP4 floating Q/DQ	0.86488	0.69728	0.76740	1.08234	0.72694	1.13178	0.73492	1.13826	0.72927	1.10384
Native encoded MXFP4	0.79393	0.98196	0.60220	1.81363	0.45672	2.34972	0.40960	2.62835	0.42983	2.58073
MXFP4 Q/DQ + MXFP8-E4M3 state	0.97213	0.22072	0.97075	0.23507	0.96922	0.22908	0.97059	0.22996	0.96850	0.23213
Flat INT4	0.63811	1.50722	0.55497	1.80492	0.50944	1.87392	0.55845	1.52630	0.52540	1.86201

with Medium confidence. No measured activity, board telemetry, matched routed BF16 design, or recurrent RTL service latency is available, so we do not derive energy per token or an energy-efficiency ratio. The matched native-MXFP8 image also fits, using 58,582 LUTs and 576 URAMs. It fails the 250-MHz check and first closes at 166.67 MHz in the tested fixed-route sweep. Its 4.952 W vectorless estimate is not measured energy.

VII. DISCUSSION

The controlled answer is no for the current implementations. Floating Q/DQ already predicts substantial four-bit drift, while

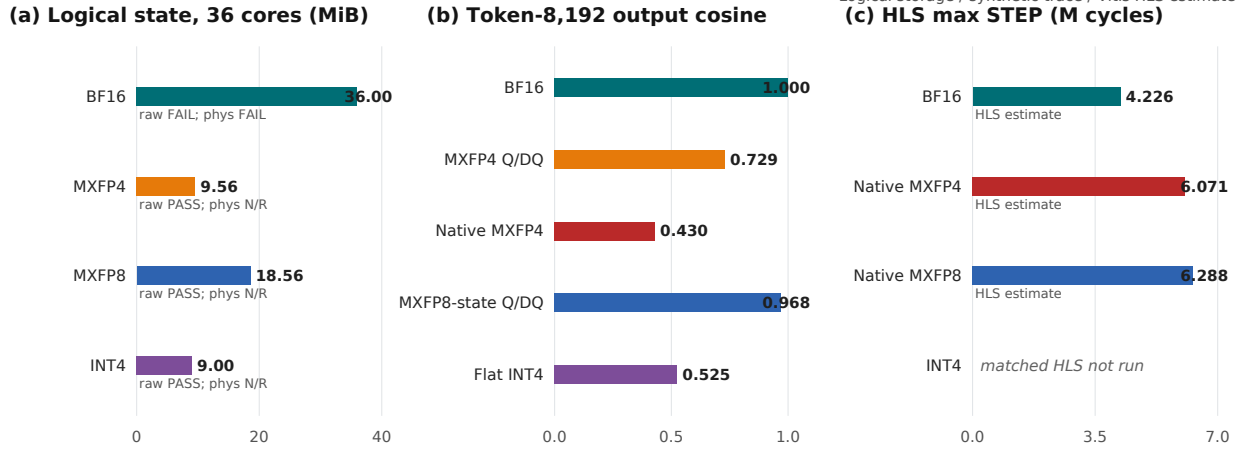
the native encoded path is worse and records 2,685,620,693 alignment underflows through token 8,192. The encoded HLS point also costs more than BF16. Adding a dense residual and recent-write log restores the finite-trace engineering criteria, but its worst longer-run state error is close to the boundary and its computation costs more than BF16.

This result separates representation density from implementation cost. Shared scales reduce bits, but exponent comparison, alignment, requantization, and folding add control and arithmetic. The E2M1 multiplier is small and DSP-free; the surrounding recurrence is not. Table IX localizes the

TABLE IV
UNIFORM-MXFP4 BLOCK-SCALE POLICIES AT TOKEN 8,192. CLIPS AND SCALE CHANGES ARE CUMULATIVE FLOATING QUANTIZE/DEQUANTIZE DIAGNOSTICS.

Scale policy	Output cosine	State rel. L_2	Clips	Scale changes
Fixed	0.847089	0.676620	67,980,414	0
Every token	0.729275	1.103841	73	26,504,191
Periodic, $N = 4$	0.773418	0.985345	16,091,670	8,345,806
Periodic, $N = 8$	0.785090	0.951738	20,026,426	4,448,732
Periodic, $N = 16$	0.785347	0.939046	22,646,994	2,354,947
Threshold triggered	0.558410	1.724816	0	35,057

Controlled evidence across three non-interchangeable levels



Raw capacity is necessary, not routed fit. Phys reports per-format implementation status. N/R points are not imputed.

Fig. 3. Evidence-level storage, quality, and schedule view, not a measured Pareto frontier. (a) Logical recurrent-state payload for 36 cores; raw PASS/FAIL is only a necessary primitive-bit capacity condition. (b) Layer-level synthetic output cosine at token 8,192; floating Q/DQ and native encoded paths remain distinct. (c) Matched fixed-geometry HLS maximum STEP estimates; the unavailable INT4 hardware point is marked not run rather than imputed.

TABLE V
FIXED-GEOMETRY U55C HLS LOGIC AND SCHEDULE ESTIMATES. MEMORY COUNTS, PACKING, AND TRANSPORT ARE EXCLUDED.

Arithmetic	Path (ns)	STEP max	LUT	FF	DSP
BF16	2.920	4,225,728	84,919	39,087	0
Native encoded MXFP4	2.920	6,070,592	138,969	43,962	3
Native MXFP8	2.920	6,287,680	143,084	44,715	3

dominant generated hierarchy. The correction's fold engine and update quantizer account for 78.7 percent of its top-level LUT estimate. A useful replacement must reduce the full state-update path, not only the encoded element width.

The word *drift* is deliberate. For one head, let E_t be candidate minus reference state. For the same external token inputs, define R_t as the exact residual between the implemented candidate transition and the ideal reference transition applied to the candidate state. This residual includes input and state quantization, alignment, accumulation, scale, clipping, and fold error. The recurrence then gives

$$E_t = \alpha_t(I - \beta_t k_t k_t^T)E_{t-1} + R_t. \quad (5)$$

Let $P_t = k_t k_t^T$. The parallel and orthogonal components obey

$$\|E_t\|_F \leq \alpha_t \|E_{t-1}\|_F + \|R_t\|_F. \quad (6)$$

TABLE VI
CORRECTED-CANDIDATE TEST AND DEVELOPMENT RESULTS. TEST VALUES ARE THE WORST ACROSS 3 SEED BLOCKS, EACH PAIRED WITH RANDOM AND ZERO INITIAL STATE. DEVELOPMENT VALUES ARE THE WORST OR TOTAL ACROSS 2 FULLY RECOMPUTED CONDITIONS. DASHES DENOTE DIAGNOSTICS NOT AGGREGATED FOR THE TEST SET.

Metric	Test (1,024)	Development (8,192)
Min. checkpoint cosine	0.994892	0.995065
Min. all-token cosine	0.994430	0.994389
Max. final state rel. L_2	0.097368	0.097472
Max. all-token state rel. L_2	0.098653	0.099057
Max. state abs. error	0.260358	0.267635
Hard-event total (three counters)	0	0
Alignment underflows	—	1,049,355,823
E2M0 residual clips	—	23,296,266
Log folds	—	2,340

For normalized k_t and $\beta_t \in [0, 1]$,

TABLE VII

LOGICAL STORAGE AND PER-LAYER HLS COMMAND ESTIMATES FOR THE RECURRENCE-AWARE MITIGATION. “AMORTIZED” CHARGES ONE FOLD EVERY 7 STEPS; IT IS NOT SERVICE P99.

Metric	BF16	Encoded MXFP4	Corrected candidate
Logical bytes/layer	1,048,576	278,528	537,744
LUT	84,919	138,969	435,089
Non-fold STEP max	4,225,728	6,070,592	17,954,144
Amortized per-layer HLS cycles/STEP	4,225,728	6,070,592	39,947,365
Fold-command burst max	–	–	153,952,544

TABLE VIII

CORRECTED-CANDIDATE OUT-OF-CONTEXT PHYSICAL AND GENERATED-RTL EVIDENCE. POWER IS VECTORLESS; THE CLOSING FREQUENCY IS THE FIRST PASSING TESTED FIXED-ROUTE POINT.

Evidence item	Result	Status
<i>Corrected MXFP4 candidate</i>		
Out-of-context physical fit	yes	PASS
250-MHz setup WNS	-1.540 ns	FAIL
First tested closing point	180.18 MHz	PASS
CLB LUT / FF	208,523 / 111,027	FIT
BRAM tile / URAM / DSP	353 / 624 / 44	FIT
DRC warnings / critical / errors	38 / 0 / 0	PASS_WITH_WARNINGS
Vectorless power at 5.6 ns	6.991 W	Medium estimate
Generated-RTL control smoke	2 / 2 commands	PASS
Candidate 64-token HLS C parity	64 tokens	PASS
Candidate generated-RTL direct LOAD	1 command	PASS
Candidate recurrent 64-token RTL parity	0 / 64 STEP commands	NOT_ESTABLISHED
<i>Native MXFP8 baseline</i>		
Out-of-context physical fit	yes	PASS
250-MHz setup WNS	-1.355 ns	FAIL
First tested closing point	166.67 MHz	PASS
CLB LUT / FF	58,582 / 39,807	FIT
BRAM tile / URAM / DSP	164 / 576 / 2	FIT
Vectorless power at 6.00 ns	4.952 W	Medium estimate

the exact identity

$$\|(I - \beta_t P_t)E\|_F^2 = (1 - \beta_t)^2 \|k_t^T E\|_2^2 + \|(I - P_t)E\|_F^2. \quad (7)$$

Thus the delta gate attenuates only error parallel to the current key; the worst-case operator norm is one because $K > 1$. Unrolling the one-step bound gives

$$\|E_T\|_F \leq \left(\prod_{i=1}^T \alpha_i\right) \|E_0\|_F + \sum_{j=1}^T \left(\prod_{i=j+1}^T \alpha_i\right) \|R_j\|_F. \quad (8)$$

If $\alpha_i \leq \bar{\alpha} < 1$ and $\|R_i\|_F \leq \bar{R}$, the right-hand side is at most $\bar{\alpha}^T \|E_0\|_F + (1 - \bar{\alpha}^T) \bar{R} / (1 - \bar{\alpha})$. Our protocol permits decay arbitrarily close to one, so it supplies no uniform contraction margin. Changing keys may attenuate different directions, but a persistent-excitation bound is not established. These relations explain why repeated local errors matter, but they are not a formal stability certificate. We do not have a conservative bound on R_t spanning input and base quantization, logged coefficients, folding, alignment, accumulation, scale changes, and clipping. “Stable” in the engineering gate therefore means only that the measured finite traces stay within its thresholds.

The correction also changes the experimental role of MXFP8. Uniform MXFP8 state is a stronger numerical baseline than MXFP4, while the corrected candidate’s logical payload is only 0.542 percent smaller. Both the corrected candidate and native MXFP8 physically fit out of context, but neither closes 250 MHz. Without measured activity, a routed BF16 comparator, or board energy, their power estimates do not establish an energy advantage for the more complex representation.

TABLE IX

GENERATED HLS HIERARCHY ABLATION. SHARES ARE RELATIVE TO EACH VARIANT’S OWN TOP-LEVEL LUT ESTIMATE AND ARE NOT ROUTED UTILIZATION.

Generated hierarchy block	LUT	Share of top (%)
BF16 state-update block	2,191	2.6
Encoded-MXFP4 state-update block	43,843	31.5
Correction fold engine	185,073	42.5
Correction update quantizer	157,419	36.2
Correction subtotal	342,492	78.7

VIII. LIMITATIONS AND THREATS TO VALIDITY

The long-horizon numerical evidence is layer-level and synthetic. The short Qwen diagnostic reconstructs query, key, value, decay, and delta-gate inputs at one pinned layer boundary, but contains only 12–18 valid tokens per prompt. It is not closed-loop execution and supports neither perplexity nor downstream-task claims. The kernel excludes projections, normalization, full-attention layers, and their key-value cache.

The nominal study uses one development seed. The correction uses a separate high-retention trace family and only 3 test seed blocks crossed with two initial states, so it is not a paired numerical ablation. Its registration was not externally timestamped with the full dependency closure; replay checks integrity, not independence. Metrics flatten heads and have no trace-population confidence interval or FP64 long-horizon curve. Thresholds are not calibrated to model quality. The longer-run error is only 0.000943 below its boundary, and changing that boundary alone to 0.09 rejects both test and development conditions. Section VII supplies no formal stability certificate.

BF16 and uniform-MXFP4 results remain HLS estimates. The correction and native MXFP8 baseline are routed out of context; both fit but fail 250 MHz. The fixed-route sweeps are not global frequency searches, and matched all-layer BF16 fails capacity before routing. Xilinx Runtime (XRT), xclbin, board checks, and synchronized same-boundary GPU measurements are absent; vectorless power is not board energy.

Packing and token quantization are excluded, and no multi-layer service latency is measured. Verification is asymmetric: uniform MXFP4 has a 64-token RTL trace; the correction has 64-token C parity, a two-command RTL control smoke, and one exact direct-RTL LOAD. Two isolated XSIM runs exhaust host memory before transaction one, and no candidate recurrent STEP completes. Results cover one U55C and one parallelism point; cross-device and longer real-trace studies remain open.

IX. CONCLUSION

Native encoded MXFP4 does not replace BF16 in this persistent-state GDN core: it has more drift and higher fixed-geometry HLS cost. A residual/write-log mitigation meets bounded synthetic criteria but exceeds BF16 cost, misses II and routed clock targets, and lacks recurrent RTL parity. The corrected and MXFP8 images fit out of context, but longer real traces, closed-loop quality, and measured board energy remain necessary.

REFERENCES

- [1] S. Yang, J. Kautz, and A. Hatamizadeh, “Gated Delta Networks: Improving Mamba2 with Delta Rule,” in *Proc. Int. Conf. Learning Representations*, 2025.
- [2] N. Gupta, P. Wang, R. Kannan, and V. K. Prasanna, “A Persistent-State Dataflow Accelerator for Memory-Bound Linear Attention Decode on FPGA,” arXiv preprint arXiv:2603.05931v1 [cs.AR], Mar. 2026, doi: 10.48550/arXiv.2603.05931. [Online]. Available: <https://arxiv.org/abs/2603.05931>
- [3] Open Compute Project, “OCP Microscaling Formats (MX) Specification,” ver. 1.0, Sep. 2023. [Online]. Available: <https://www.opencompute.org/documents/ocp-microscaling-formats-mx-v1-0-spec-final-pdf>
- [4] Qwen Team, “Qwen3-Next-80B-A3B-Instruct,” Hugging Face model repository, rev. 9c7f2fbe84465e40164a94cc16cd30b6999b0cc7, accessed Aug. 1, 2026. [Online]. Available: <https://huggingface.co/Qwen/Qwen3-Next-80B-A3B-Instruct/tree/9c7f2fbe84465e40164a94cc16cd30b6999b0cc7>
- [5] Hugging Face, “Qwen3-Next implementation,” *Transformers*, ver. 4.57.0, commit 8ac2b916b042b1f78b75c9eb941c0f5d2cdd8e10, 2025. [Online]. Available: https://github.com/huggingface/transformers/blob/8ac2b916b042b1f78b75c9eb941c0f5d2cdd8e10/src/transformers/models/qwen3_next/modeling_qwen3_next.py
- [6] FLA Contributors, “Gated delta rule implementation,” *flash-linear-attention*, ver. 0.2.1, commit a670dff4c2537fc1a82486584dd9569e18fba833, 2025. [Online]. Available: https://github.com/fla-org/flash-linear-attention/tree/a670dff4c2537fc1a82486584dd9569e18fba833/fla/ops/gated_delta_rule
- [7] S. Yang, B. Wang, Y. Zhang, Y. Shen, and Y. Kim, “Parallelizing Linear Transformers with the Delta Rule over Sequence Length,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024, doi: 10.52202/079017-3668.
- [8] H. Wen, Z. Wang, J. Dong, W. Lou, L. Gong, C. Wang, and X. Zhou, “QLlama: An FPGA-Based Microscaling Quantization Accelerator for Energy-Efficient Llama2 Inference,” *IEEE Embedded Systems Letters*, vol. 17, no. 5, pp. 337–340, 2025, doi: 10.1109/LES.2025.3600563.
- [9] M. Mekhemer, A. Elsousy, B. Venkatesh, R. Rowley, V. Betz, N. Kapre, and A. Boutros, “Jack of All Scales: A Versatile FPGA Tensor Block for MXFP Precisions,” arXiv preprint arXiv:2607.13898v1 [cs.AR], Jul. 2026, doi: 10.48550/arXiv.2607.13898.
- [10] G. Karfakis *et al.*, “MXFormer: A Microscaling Floating-Point Charge-Trap Transistor Compute-in-Memory Transformer Accelerator,” arXiv preprint arXiv:2602.12480v1 [cs.AR], Feb. 2026, doi: 10.48550/arXiv.2602.12480.
- [11] W. Liu, H. Meng, Y. Luo, P. Zhang, and X. Ma, “MicroMix: Efficient Mixed-Precision Quantization with Microscaling Formats for Large Language Models,” in *Proc. Int. Conf. Learning Representations*, 2026.
- [12] J. Yu *et al.*, “MXAttention: Data-Free Optimal Scaling and Pre-Normalization Quantization for MXFP4 Attention,” arXiv preprint arXiv:2607.24377v1, Jul. 2026, doi: 10.48550/arXiv.2607.24377.
- [13] M. Zhang *et al.*, “Benchmarking Post-Training Quantization of Large Language Models under Microscaling Floating Point Formats,” arXiv preprint arXiv:2601.09555v1, Jan. 2026, doi: 10.48550/arXiv.2601.09555.
- [14] L. Zhang *et al.*, “When Good Enough Is Optimal: Multiplication-Only Matrix Inversion Approximation for Quantized Gated DeltaNet,” in *Proc. 43rd Int. Conf. Machine Learning*, PMLR, vol. 306, 2026.
- [15] Z. Liu, J. Yuan, H. Jin, S. Zhong, Z. Xu, V. Braverman, B. Chen, and X. Hu, “KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache,” in *Proc. 41st Int. Conf. Machine Learning*, PMLR, vol. 235, pp. 32332–32344, 2024.
- [16] H. Kim, B. Ko, M. Kang, M. Kim, D. Lee, J. Lee, S. Yoon, and D. Jung, “SSDi8: Accurate and Efficient 8-bit Quantization for State Space Duality,” in *Proc. Int. Conf. Learning Representations*, 2026.